River Ice Segmentation with Deep Learning Supplementary

Abhineet Singh Hayden Kalke Mark Loewen Nilanjan Ray
University of Alberta
asingh1,kalke1,mrloewen,nray1@ualberta.ca

I. Introduction

This supplementary provides results on additional unlabeled images (Fig. 3, 4), details of tested video sequences (Table II) and categorized links for Google photos albums showing results on labeled images (Table I) and Google Drive folders containing video results (Table III). Each row in the albums shows (from left to right) the original image, its ground truth label and its segmentation result. All these results are also available in the accompanying data [1].

Results comparing different backbone networks with DeepLab are also included in Sec. II.

TABLE I: Google Photos albums showing labeled qualitative results for ablation tests

Model	n_training_images					n_pixels_per_class			
SVM	32	24	16	8	4	-			
Deeplab	32	24	16	8	4	1000	100	10	2
UNet	32	24	16	8	4	1000	100	10	2
SegNet	32	24	16	8	4	1000	100	10	2
DenseNet	32	24	16	8	4	1000	100	10	2

TABLE II: Details of the 5 video sequences used for testing

ID	Name	Frame Count	Start Frame	End Frame
1	20160121 / YUN00001	3600	1	3600
2	20160122 / YUN00002	1800	701	2500
3	20160122 / YUN00020	1800	2001	3800
4	20161203 / Deployment_1_YUN00001	1800	901	2700
5	20161203 / Deployment_1_YUN00002	1800	1	1800

TABLE III: Google Drive links for video results

Type	Video ID					Type	Video ID				
deeplab_densenet_unet						**selective_pixel_ablation**					
combined	1	2	3	4	5	**densenet**	1	3			
frazil	1	2	3	4	5	**deeplab**	1	3			
anchor	1	2	3	4	5	**unet**	1	3			
svm_densenet_unet						**compare**	1				
combined	1	3	4			**training_images_ablation**					
frazil	1	3	4			**densenet**	1	2	3	4	5
anchor	1	3	4			**deeplab**	1	2	3	4	5

II. Comparing DeepLab Backbone Networks

This section provides results comparing three backbone networks for DeepLab - Xception65 (used in the main paper) [2], Auto-DeepLab [4] and a modified version of ResNet101 [3] where the first 7×7 convolution has been replaced by three 3×3 convolutions similar to PSPNet [5].

A. Overview

Table IV shows the results for both segmentation recall and precision as well as ice concentration median mean absolute error (MAE). It can be seen that Xception65 has similar performance as the other newer backbones and is even slightly better in most metrics except segmentation recall where Auto-DeepLab and ResNet101 (PSP) dominate for anchor and frazil ice respectively.

TABLE IV: Segmentation recall and precision along with ice concentration median MAE for DeepLab with three different backbone architectures trained and tested on the 32 and 18 image sets respectively.

Backbone	anchor ice	frazil ice	ice+water	ice+water(fw)	Median MAE (%)
Recall (%)					**Anchor Ice**
Xception65	74.46	87.51	**86.38**	**90.87**	**4.71**
Auto-Deeplab	**78.69**	81.28	85.70	90.21	5.06
ResNet101 (PSP)	67.07	**93.24**	85.16	90.81	4.78
Precision (%)					**Frazil Ice**
Xception65	**62.39**	**77.14**	**77.25**	**84.32**	**4.52**
Auto-Deeplab	59.44	73.43	75.57	83.66	5.24
ResNet101 (PSP)	59.79	76.47	76.34	84.01	5.49

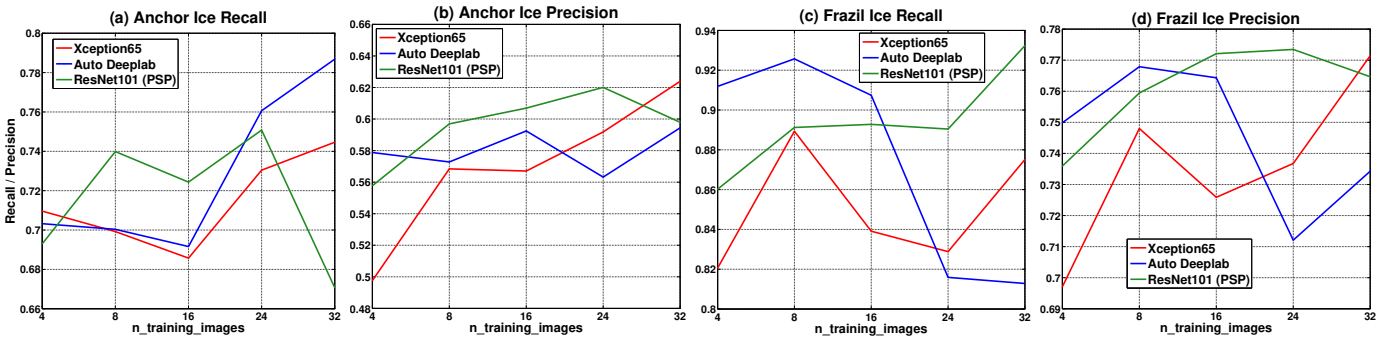

Fig. 1: Results of ablation tests with training images for (a-b) anchor ice and (c-d) frazil ice. Note the variable Y-axis limits.

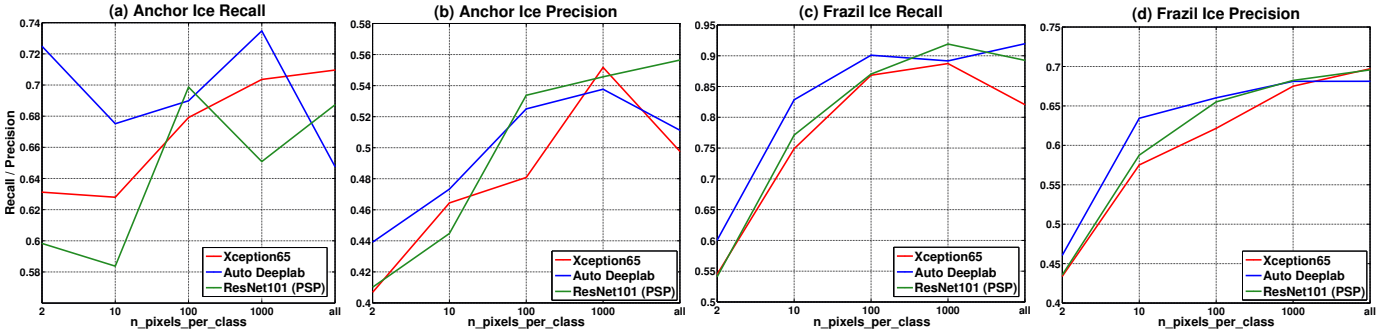

Fig. 2: Results of ablation tests with selective pixels for (a-b) anchor ice and (c-d) frazil ice. Note the variable Y-axis limits.

B. Ablation Tests

Fig. 1 and 2 respectively show the results of training images and selective pixel ablation tests. As in the previous section, Xception65 remains competitive with the other backbones though it seems to be slightly more outperformed here especially with fewer training images in Fig. 1 and frazil ice selective pixel tests in Fig. 2.

References

[1] Alberta River Ice Segmentation Dataset. http://dx.doi.org/10.21227/ebax-1h44, 2019.
[2] F. Chollet. Xception: Deep learning with depthwise separable convolutions. In *CVPR*, pages 1800–1807, 2017.
[3] K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. In *2016 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2016, Las Vegas, NV, USA, June 27-30, 2016*, pages 770–778, 2016.
[4] C. Liu, L.-C. Chen, F. Schroff, H. Adam, W. Hua, A. Yuille, and L. Fei-Fei. Auto-deeplab: Hierarchical neural architecture search for semantic image segmentation. In *CVPR*, 2019.
[5] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia. Pyramid scene parsing network. In *Proceedings of IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.

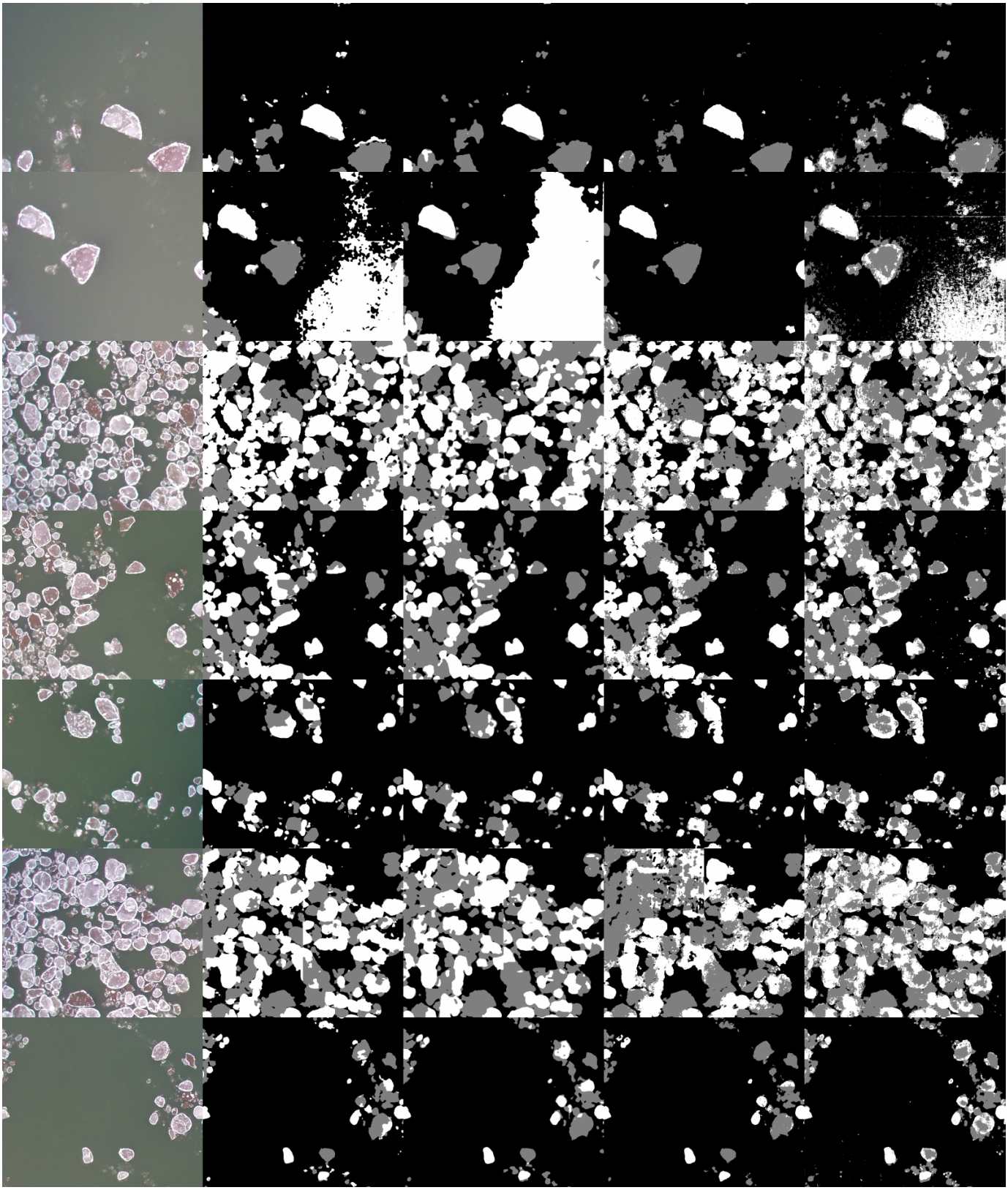

Fig. 3: Results of applying the best configurations of the four models on unlabeled images: left to right: Raw image, UNet, SegNet, Deeplab, DenseNet

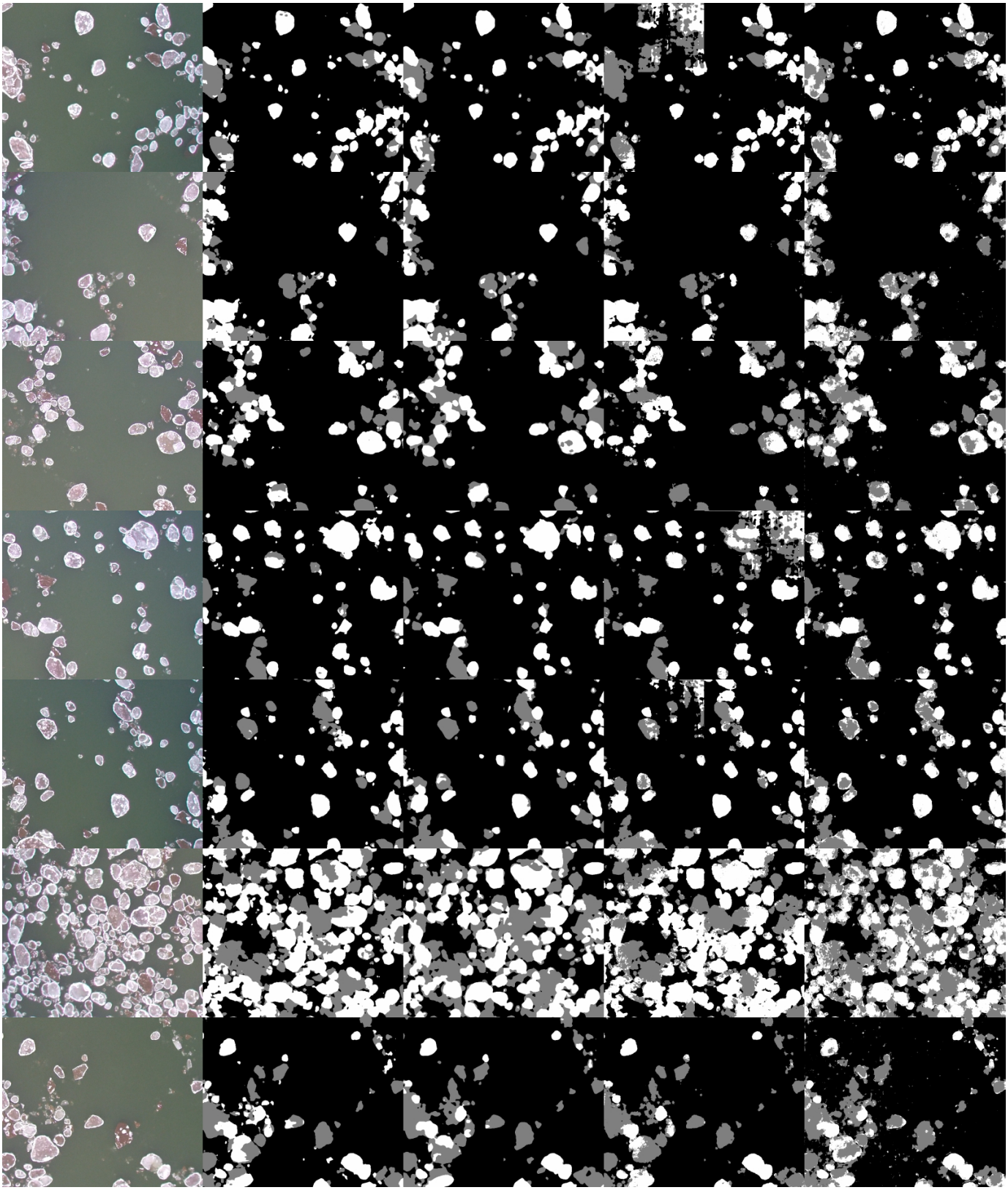

Fig. 4: Results of applying the best configurations of the four models on unlabeled images: left to right: raw image, UNet, SegNet, Deeplab, DenseNet